

Logic Puzzles and Variables Mission

Warm-up and worked example

Name: _____

Date: _____

PAGE 1 | WEEK 9, DAY 2

YOUR MISSION Use organized reasoning, elimination, and variables to solve logic problems.

SMART MOVE Draw a simpler case and organize what you know.

Math Talk

If $a + b = 20$ and a is 4 more than b , what could you try first?

My idea:

New Words

represent	deduce	assumption
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Solve a small three-person logic puzzle.
2. Record clues in a table.

What the example shows: _____

Logic Puzzles and Variables Mission

Learn it and show your thinking

Name: _____

Date: _____

PAGE 2 | WEEK 9, DAY 2

YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Solve a small three-person logic puzzle.

[] **Step 2:** Record clues in a table.

[] **Step 3:** Cross out impossibilities.

[] **Step 4:** Translate a clue into an equation.

[] **Step 5:** Check that the final answer satisfies every clue.

My model, drawing, or organized work:

The big idea in my own words:

Logic Puzzles and Variables Mission

Practice lab

Name: _____

Date: _____

PAGE 3 | WEEK 9, DAY 2

YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. One ordering puzzle.

2. One missing-number puzzle.

3. One balance puzzle.

4. One cryptarithm with distinct digits.

Choose one answer and prove it another way:

Logic Puzzles and Variables Mission

Challenge and final check

Name: _____

Date: _____

PAGE 4 | WEEK 9, DAY 2

YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Use different digits for A and B: $AB + BA = 99$. Find all solutions.

My plan:

My solution and proof:

Final Check

State one fact you knew directly and one fact you deduced.

☐ I can teach it

☐ I need practice

☐ I need help

Logic Puzzles and Variables Mission

Reusable math tool

Name: _____

Date: _____

PAGE 5 | WEEK 9, DAY 2

YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Draw a simpler case and organize what you know.

Problem-Solving Organizer

What I know

What I need to find

My model or table

My equations

My check and explanation
